F72-SR-001
415V REAGENTS SWITCHROOM

F72-SW-072
415V REAGENTS 415V MOTOR CONTROL CENTRE

CONTINUED ON SHT 2
2000-F72-SLD-EL-10202

CONTINUED ON SHT 4
2000-F72-SLD-EL-10204

NORTHERN STAR

Approved	[X]
Approved as Noted	[ ]
Not Approved	[ ]
For Information Only	[ ]

Signed [redacted] Date 10/12/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

415V 3NE~ 50Hz 2500A 65kA for 1sec

	4R-21	3F-05	3F-09	3F-13	3F-17	3F-21	4F-01	4F-05	4F-09	4F-13	1R-07	2R-01
	160A	50A	50A	50A	63A	63A	20A	20A	32A	32A	100A	100A
	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A
	100/1A	70A	70A	70A	70A	70A	100/1A	100/1A	70A	70A	70A	70A
		EOL	EOL	EOL	EOL	EOL			EOL	EOL	EOL	EOL
	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~
	55kW	15kW	15kW	15kW	22kW	22kW	1.5kW	1.5kW	7.5kW	7.5kW	30kW	30kW
	F72-MB-412F LIME SLAKING MILL 2 MOTOR 6 2000-F72-SCM-EL-10681	F72-PP-424 CIL4 DILUTED ACID DOSING PUMP 2000-F72-SCM-EL-10450	F72-PP-673 CAUSTIC RINGMAIN PUMP 1 2000-F72-SCM-EL-10401	F72-PP-674 CAUSTIC RINGMAIN PUMP 2 2000-F72-SCM-EL-10402	F72-PP-675 CYANIDE RINGMAIN PUMP 1 2000-F72-SCM-EL-10501	F72-PP-676 CYANIDE RINGMAIN PUMP 2 2000-F72-SCM-EL-10502	F72-PP-681 LEAD NITRATE DOSING PUMP 1 2000-F72-SCM-EL-10551	F72-PP-682 LEAD NITRATE DOSING PUMP 2 2000-F72-SCM-EL-10552	F72-PP-683 ACID DOSING PUMP 1 2000-F72-SCM-EL-10451	F72-PP-684 ACID DOSING PUMP 2 2000-F72-SCM-EL-10452	F72-PS-419 MILK OF LIME AREA SUMP PUMP 1 2000-F72-SCM-EL-10659	F72-PS-466 MILK OF LIME AREA SUMP PUMP 2 2000-F72-SCM-EL-10666

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F72-SW-072 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F72-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- JB — JUNCTION BOX
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

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F72-SW-072 415V MCC - single line diagram sheet 3

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